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Cell cycle in mouse development

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Mice likely represent the most-studied mammalian organism, except for humans. Genetic engineering in embryonic stem cells has allowed derivation of mouse strains lacking particular cell cycle proteins. Analyses of these mutant mice, and cells derived from them, facilitated the studies of the functions of cell cycle apparatus at the organismal and cellular levels. In this review, we give some background about the cell cycle progression during mouse development. We next discuss some insights about *in vivo* functions of the cell cycle proteins, gleaned from mouse knockout experiments. Our text is meant to provide examples of the recent experiments, rather than to supply an extensive and complete list.
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Cell cycles in early development

Mouse embryo development starts when highly specialized haploid gametes, oocytes and sperm cells meet during the process of fertilization. Prior to fertilization, an oocyte grows and accumulates factors necessary to sustain early embryonic development (Sorensen and Wassarman, 1976; Eppig, 1996). Indeed, one-cell stage mouse embryo depends exclusively on maternal protein stocks. Initial transcription from the embryonic genome occurs during the G2 phase of the first embryonic cell cycle (Bouniol *et al*., 1995) and soon, at the two-cell stage, it is followed by 'embryonic genome activation', that is, a burst of new mRNA synthesis followed by protein synthesis (Flach *et al*., 1982). This process is accompanied by substantial degradation of factors accumulated during oocyte growth. However, some of the maternally inherited transcripts and proteins are detectable even at the end of the preimplantation development, that is, at a blastocyst stage (Bachvarova and De Leon, 1980).

Somatic and early cell cycles of the mouse embryo seem to progress according to the same general scheme. For instance, the lengths of preimplantation and somatic cell cycles are comparable. Since mouse embryo needs 72 h to reach a blastocyst stage ($\sim$64 cells), the mean cell division time during this period was calculated to be approximately 14 h (Hogan *et al*., 1994). However, the two first cell cycles are longer and each of them takes 18–20 h. The first cell cycle starts when ovulated oocyte becomes fertilized by sperm. This event triggers a series of morphological and biochemical changes including transformation of female and male chromatin into functional nuclei. The oocyte completes the second meitotic division, its chromosomes decondense and the so-called female pronucleus is formed. Simultaneously, highly condensed sperm chromatin undergoes a sequence of structural changes (including replacement of sperm-specific protamines by histones) and transforms into haploid male pronucleus (Wright and Longo, 1988; Adenot *et al*., 1991; Yanagimachi, 1994). Formation of both pronuclei takes approximately 2 h. DNA replication starts simultaneously in female and male pronuclei 8–9 h after fertilization. The first embryonic S phase lasts approximately 6 h and it is followed by a 4-h long G2 phase (Howlett and Bolton, 1985; Adenot *et al*., 1991; Bouniol-Baly *et al*., 1997). Both sets of parental chromatin remain physically separated until the time of the first mitotic division (Donahue, 1972; Howlett and Bolton, 1985; Mayer *et al*., 2000). Finally, at the end of the first cell cycle, pronuclei concurrently undergo nuclear envelope breakdown, and their condensing chromosomes align within the common metaphase plate of the first mitotic division (Adenot *et al*., 1991; Mayer *et al*., 2000).

The second cell cycle is as long as the first one. However, the second S-phase starts almost immediately after the first mitosis is completed, with the G1 phase lasting only 1 h (Bolton *et al*., 1984; Ciemerych *et al*., 1999). Duration of the third (four-cell stage) and fourth (eighth-cell stage) cell cycles is shorter, each of them taking approximately 12 h (Smith and Johnson, 1986). Moreover, the fourth cleavage division (8–16 cells) results in the generation of blastomeres that differ in size and that will later divide with different dynamics. Thus, the smaller cells complete the fifth cleavage after 14 h, while bigger ones take only 12 h to divide (MacQueen and Johnson, 1983). More pronounced differences in the dynamics of cellular proliferation are observed at a time when preimplantation mouse embryo reaches a blastocyst stage. At this point, two cellular lineages are easily distinguishable: inner cell mass that will give rise to the embryo proper, and trophectoderm that will form extraembryonic tissues. Cells of the polar

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trophectoderm – a part of the trophectoderm that remains in close contact with the inner cell mass and gives rise to the ectoplacental cone – divide more vigorously than cells composing mural trophectoderm (Copp, 1978). After implantation, at the onset of gastrulation, various lineages continue to proliferate with different kinetics. For instance, the doubling time of embryonic ectoderm cells was estimated to be 5 h at day 6.5 of development, and 8 h at day 7.5 (Snow, 1977; Lawson *et al*., 1991). The doubling time of embryonic mesoderm is 14 h at day 7.5 of development (Snow, 1977). In contrast, during this period, cells within the so-called 'proliferative zone' localized at the anterior end of the primitive streak undergo mitotic divisions every 2–3 h (Snow, 1977; Mac Auley *et al*., 1993). Cells of this rapidly dividing subpopulation dramatically shorten their G1 and G2 phases to the point that they effectively eliminated the two 'gap' phases. In addition, the length of the S phase is also reduced (Mac Auley *et al*., 1993). The above observations suggest the existence of the mechanisms specific for these early embryonic cell cycles.

Later in development, during organogenesis, the length of the cell cycles increases again and becomes similar to those of most proliferating adult tissues. Nevertheless, variable growth dynamics of embryonic tissues and organs is observed during the whole prenatal development (Goedbloed, 1977).

The available data do not provide satisfactory explanation for the molecular basis of changes in proliferation dynamics during pre- and peri-implantation mouse development. However, several important observations about the core cell cycle machinery during these early embryonic cycles have been reported. Thus, the majority of proteins deemed responsible for the regulation of G1, S, G2 and M phases are present in preimplantation mouse embryos (Choi *et al*., 1991; Moore *et al*., 1996; Palena *et al*., 2000; Kohoutek *et al*., 2004). However, some observations questioned the importance of certain cell cycle regulators during the early cell cycles. For example, preimplantation development was reported to progress independently of the exogenous mitogens, that is, regardless of the presence of growth factors in the environment (Biggers *et al*., 1997; Stewart and Cullinan, 1997). These observations may suggest that the regulatory pathways that are triggered by the mitogenic stimulation, such as cyclin D–CDK→retinoblastoma protein pathway, may not operate during this period of embryogenesis. It should be noted, however, that the use of knockout mice to define cell cycle components that are essential during early embryogenesis is confounded by the presence of maternally derived factors in the embryos.

In vivo functions of cell cycle regulators

D-cyclins, CDK4 and CDK6

Three D-type cyclins – cyclin D1, D2 and D3 – operate in mammalian cells. D-cyclins bind their catalytic partners – cyclin-dependent kinases CDK4 and CDK6 – and phosphorylate members of the retinoblastoma (pRb) family. This in turn leads to release of E2F transcription factors and results in the activation of transcription of E2F-responsive genes (Sherr and Roberts, 1999, 2004).

All the three D-type cyclins were shown to be present in unfertilized mouse oocytes (Moore *et al*., 1996; Kohoutek *et al*., 2004); cyclin D3 levels decrease sharply during the first cell cycle (Kohoutek *et al*., 2004). Similarly, the levels of the catalytic partners for D-type cyclins, CDK4 and CDK6 rapidly decrease at the beginning of mouse embryo development (Kohoutek *et al*., 2004). Moreover, the major target of cyclin D-CDK4/6 complexes – the retinoblastoma protein- is drastically downregulated at the two-cell stage and remains absent until the late blastocyst stage (Moore *et al*., 1996; Iwamori *et al*., 2002). The absence of pRb in early embryonic cells would be predicted to result in a constitutive activation of E2F transcription factors, even without cyclin D/CDK4/6 activity. Consistent with the notion that the cyclin D-CDK4/6→pRb pathway is not functional at this stage are the observations that mouse embryonic stem cells are resistant to cyclin D-CDK4/6-specific inhibitor – $p16^{INK4a}$ (Savatier *et al*., 1996). However, recent report indicates that mouse embryonic stem cells express cyclin D3–CDK6 complexes, and that these complexes are resistant to the inhibition by $p16^{INK4a}$ (Faast *et al*., 2004). Hence, cyclin D-dependent kinases may play a role in cell cycle control of the early embryos.

During later stages of mouse embryo development, cyclins D are expressed in a tissue-specific, but greatly overlapping manner (Aguzzi *et al*., 1996; Wianny *et al*., 1998; Ciemerych *et al*., 2002).

Mice lacking individual D-type cyclins have been generated. Although the expression of each of the D-cyclins is quite widespread, cyclin D-deficient mice displayed very narrow phenotypes (Table 1). Thus, cyclin D1-null mice were viable and displayed neurological abnormalities and hypoplastic retinas. Moreover, the mammary glands of cyclin D1-null females failed to undergo normal lobuloalveolar development during pregnancy (Fantl *et al*., 1995; Sicinski *et al*., 1995). Cyclin D2-null females were sterile due to the proliferative failure of ovarian granulosa cells, while the males were fertile but displayed hypoplastic testes (Sicinski *et al*., 1996). Moreover, cerebellar development (Huard *et al*., 1999), adult neurogenesis (Kowalczyk *et al*., 2004), B-lymphocyte proliferation (Lam *et al*., 2000; Solvason *et al*., 2000) and pancreatic β-cell proliferation (Georgia and Bhushan, 2004; Kushner *et al*., in press) were affected in the absence of cyclin D2. Cyclin D3-deficient mice displayed a defect in the development of T-lymphocytes (Sicinska *et al*., 2003).

Also, mice lacking combinations of the D-type cyclins have been generated. These double knockout mice displayed phenotypes representing the sum of the abnormalities seen in mice lacking individual cyclins (Ciemerych *et al*., 2002). However, no new phenotypes were encountered (Figure 1, Table 1). These observations suggested that either the remaining, intact D-cyclin allowed relatively normal development, or, alternatively, the proliferation of several cell types could occur in a cyclin D-independent fashion. Recent generation and characterization of mice lacking all the three D-cyclins unexpectedly supported the latter possibility (see below).

Concurrently with these studies, mice lacking catalytic partners of the D-type cyclins, CDK4 and CDK6, were also characterized. Both CDK4- and CDK6-deficient mice were viable and displayed abnormalities within the similar compartments to those affected by the deficiency of D-cyclins. Thus, CDK4-null mice were smaller (like cyclin D1-deficient mice) and also showed defects resembling those encountered in cyclin D2-null animals (ovarian and testicular defects, pancreatic β-cell hypoplasia) (Rane *et al.*, 1999; Tsutsui *et al.*, 1999; Mettus and Rane, 2003). Moreover, mice lacking CDK6 displayed hematopoietic defects resembling abnormalities seen in cyclin D-deficient mice (Malumbres *et al.*, 2004). Furthermore, mice lacking Bmi-1, a repressor of p16^{INK4a}, displayed hypoplastic cerebella and suffered from hematopoietic abnormalities, again resembling cyclin D-deficient animals (van der Lugt *et al.*, 1994; Jacobs *et al.*, 1999; Park *et al.*, 2003; Leung *et al.*, 2004).

Recently, mice lacking all three D-type cyclins, or lacking CDK4 plus CDK6 have been described. Kozar *et al.* (2004) demonstrated that advanced stages of mouse embryo development can be reached without any of the D-type cyclins. Cyclin D1$^{-/-}$D2$^{-/-}$D3$^{-/-}$ embryos developed relatively normally until mid-gestation, and died prior to day 17.5 of pregnancy (the entire mouse gestation takes 19 days). Detailed analyses of cyclin D-null embryos and fibroblasts derived from them revealed that the majority of cell types could proliferate relatively normally in the absence of the D-type cyclins. The proliferative failure was limited to myocardial cells and to hematopoietic stem cells, demonstrating that these lineages critically require D-type cyclins (Kozar *et al.*, 2004).

Very similar phenotypes were observed in mice lacking CDK4 and CDK6. These mice survived few days longer, likely due to an ability of the D-cyclins to associate with CDK2 in CDK4$^{-/-}$CDK6$^{-/-}$ tissues (Malumbres *et al.*, 2004). Collectively, the knockouts of the D-cyclins, CDK4 and CDK6, provided genetic evidence that the major function of the D-cyclins in the developing embryo is to activate CDK4 and CDK6. Moreover, these studies revealed that the proliferation of only selected compartments requires cyclin D–CDK activity (hematopoietic cells, myocardium and tissues affected in cyclin D1, D2, D3, CDK4 or CDK6 knockouts). One of the major challenges in the field will be to dissect the differences between the cell cycle machinery in cyclin D-dependent versus cyclin D-independent cell types.

Mice lacking D-cyclins or CDK4 were shown to be resistant to certain malignancies (Table 2). Cyclin D1-deficient mice were resistant to breast cancers driven by the activated *Ras* and *c-ErbB-2* oncogenes, while being fully susceptible to mammary carcinomas triggered by the Myc and Wnt-1 (Yu *et al.*, 2001). In addition, cyclin D1-null mice showed reduced susceptibility to Ras-mediated skin tumorigenesis (Robles *et al.*, 1998), and to intestinal polyps in the Apc^{Min} background (Hulit *et al.*, 2004). CDK4-deficient mice, in turn, showed reduced susceptibility to Myc-induced tumors in the oral mucosa (Miliani De Marval *et al.*, 2004). Collectively, these findings raise the possibility that cyclin D–CDK complexes might represent potential therapeutic targets in human cancers. It remains to be demonstrated, however, whether cyclin D–CDK complexes are required not only for the initiation of tumors, but also for the maintenance of tumor cell proliferation.

The retinoblastoma protein, p107 and p130

The retinoblastoma protein, pRb, along with pRb-related proteins p107 and p130 represent the major targets for cyclin D–CDK4/6 complexes (Sherr and Roberts, 1999, 2004).

The pRb family members, also referred to as 'pocket proteins', are differentially expressed during mouse embryo development. pRb is absent at the preimplantation stages, but it becomes upregulated at the time of implantation, that is, at a blastocyst stage (Iwamori *et al.*, 2002). Expression patterns of p107 and p130 at preimplantation stages were not determined so far. Later in development, some selected tissues express only a single member of the pRb family. For instance, at day 13.5 of embryonic development, muscle, lens and retinas express exclusively pRb, while the heart, lung, kidney and intestines express only p107. Yet other compartments, such as nervous system or liver, express all pRb-like proteins (Jiang *et al.*, 1997).

Mice lacking pRb died *in utero* at midgestation (12–15 day of pregnancy) due to a severe anemia. Mutant embryos displayed defects in lens development, massive cell death within the central nervous system (CNS) and peripheral nervous system (PNS), as well as abnormal S-phase entry of post-mitotic neurons (Clarke *et al.*, 1992; Jacks *et al.*, 1992; Lee *et al.*, 1992; Morgenbesser *et al.*, 1994). These findings strongly suggested a critical role for pRb in embryo development. Surprisingly, analyses of mouse chimeras composed of wild-type and pRb-deficient cells revealed that the mutant cells were able to significantly contribute to several adult tissues including the erythroid lineage. However, chimeric animals displayed an increased number of immature (nucleated) erythrocytes during fetal erythropoiesis. Moreover, neuronal apoptotic defects were not observed in pRb-deficient cells within these chimeras, suggesting that this phenotype was not cell autonomous (Maandag *et al.*, 1994; Williams *et al.*, 1994b). On the other hand, extensive ectopic S-phase entry persisted in the brains of chimeric embryos, implying that this phenotype might be cell autonomous (Lipinski *et al.*, 2001). This notion was further tested by generation of mice with *Rb* gene conditionally targeted within the CNS, PNS and lens (MacPherson *et al.*, 2003) or within the telencephalon (Ferguson *et al*., 2002). These animals underwent embryonic development in the presence of normal erythropoiesis and were hence able to survive until birth. Mutant mice displayed ectopic S-phase entry, but no apoptosis within the CNS (Ferguson *et al*., 2002; MacPherson *et al*., 2003). Since hypoxia-inducible genes were shown to be induced within the CNS of pRb-deficient embryos, it was suggested that apoptosis observed in mutant mice might have been caused by hypoxia due to defects in erythropoiesis (MacPherson *et al*., 2003).

Recently, a role of pRb in the erythroid lineage was supported by *in vitro* erythroid differentiation culture experiments. These analyses revealed that the loss of pRb in differentiating erythroid cells resulted in impaired cell cycle exit and impaired terminal differentiation (Clark *et al*., 2004). Moreover, it was recently shown that pRb is necessary to promote normal erythroblast expansion and red cell enucleation under stress conditions (Spike *et al*., 2004).

In a truly stunning development, Wu *et al*. (2003) demonstrated that many phenotypes of pRb-deficiency could be ascribed to placental abnormalities. The authors found that abnormal expansion and differentiation of trophoblast cells caused the failure of the labyrinth development in pRb-deficient placentas, resulting in improper nutrient and oxygen exchange between the mother and the embryo (Wu *et al*., 2003). To circumvent the placental abnormalities, the authors generated mice composed of pRb-null embryos and wild-type placentas using 'tetraploid complementation' technique. pRb-deficient embryos developed to term, but died due to respiratory problems. Importantly, developmental abnormalities that were thought to cause embryonic lethality in $pRb^{-/-}$ mice were largely absent. Thus, the development of the erythroid lineage was almost entirely rescued. A small fraction of nucleated erythrocytes was present, however, suggesting a mild delay in erythroid differentiation. Thus, no visible abnormalities in the development of the nervous system were detected. Surprisingly, longer survival of pRb mutant embryos resulted in the manifestation of the defects in the skeletal muscle development (Wu *et al*., 2003). Similar defects of myogenesis were observed in mice carrying the hypomorphic *Rb* minigene, and hence expressing low levels of pRb (Zacksenhaus *et al*., 1996).

Importantly, phenotypes of pRb deficiency can also be rescued by ablation of several cell cycle regulators, thereby providing a genetic evidence for their interaction with pRb. For instance, ablation of downstream targets of pRb – E2F1 and E2F3 transcription factors – resulted in prolonged development of mutant embryos. Double mutant conceptuses were able to survive until E17.0–E17.5, most likely due to significant, but not full, rescue of the erythroid defects (Tsai *et al*., 1998; Ziebold *et al*., 2001). Apoptosis and ectopic S-phase entry in the CNS were also reduced in $pRb^{-/-}E2F1^{-/-}$ and $pRb^{-/-}E2F3^{-/-}$, but not in $pRb^{-/-}E2F2^{-/-}$ embryos (Tsai *et al*., 1998; Ziebold *et al*., 2001; Saavedra *et al*., 2002). Interestingly, cell death in the PNS was alleviated by ablation of E2F3 but not E2F1, suggesting that the latter factor is not involved in the induction of apoptosis in this compartment (Tsai *et al*., 1998; Ziebold *et al*., 2001). Moreover, defects in muscle development become manifested in $pRb^{-/-}E2F1^{-/-}$ and $pRb^{-/-}E2F3^{-/-}$ animals (Tsai *et al*., 1998; Ziebold *et al*., 2001).

Also ablation of Id2, another pRb-associated protein, prolonged the embryonic lifespan of pRb-deficient embryos, and led to the manifestation of defects in myogenesis. Notably, ablation of Id2 rescued several phenotypes of pRb-deficiency, such as ectopic S-phase entry and increased apoptosis within the nervous system, as well as hematopoietic abnormalities (Lasorella *et al*., 2000). It should be noted that Id2 was shown to play a role during human placental development (Janatpour *et al*., 2000), suggesting a possibility that at least some of the aspects of the observed rescue can be attributed to the genetic interaction between pRb and Id2 within the placentas.

As described above, analyses of mice lacking pRb revealed an important role of pRb in muscle differentiation. Consistent with these observations are earlier findings that pRb cooperates with MyoD transcription factor and participates in the regulation of myogenesis (Novitch *et al*., 1999). The activity of MyoD was also shown to be influenced by the members of the Ras family (Kong *et al*., 1995; Ramocki *et al*., 1998). Intriguingly, analyses of mice lacking both pRb and N-Ras revealed that the loss of N-Ras led to a significant restoration of skeletal muscle development in pRb-deficient mice (Takahashi *et al*., 2003). These findings point to a genetic interaction between pRb and N-Ras in controlling muscle differentiation.

Mice lacking other pRb family members – p107 or p130 – displayed no obvious abnormalities, suggesting that in most mouse tissues these proteins play redundant functions (Cobrinik *et al*., 1996; Lee *et al*., 1996). However, this stands to be true only for mutants generated in mixed 129/Sv × C57BL/6 genetic background. A dramatically different phenotype was observed when knockout animals were generated in BALB/cJ background. These p107-deficient mice displayed pronounced growth retardation, myeloid hyperplasia in spleens and livers, while *in vitro* cultured p107-null fibroblasts and myoblasts showed accelerated proliferation (LeCouter *et al*., 1998a). p130-deficient embryos, generated in BALB/cJ background died *in utero* between days 11 and 13 of development due to defects in neurogenesis (increased apoptosis and proliferation, decreased number of differentiated neurons) and myogenesis (LeCouter *et al*., 1998b). Importantly, a single backcross with C57BL/6 mice corrected all the observed phenotypes postulating the existence of a modifier in BALB/cJ mice. Reduced $p16^{INK4a}$ activity observed in this particular strain of mice might be responsible for the severe phenotype of $p107^{-/-}$ or $p130^{-/-}$ deficiencies (Zhang *et al*., 2001). Double knockout $p107^{-/-}p130^{-/-}$ mice, generated in the mixed 129/Sv × C57BL/6 genetic background, died immediately after birth, indicating that the two proteins play

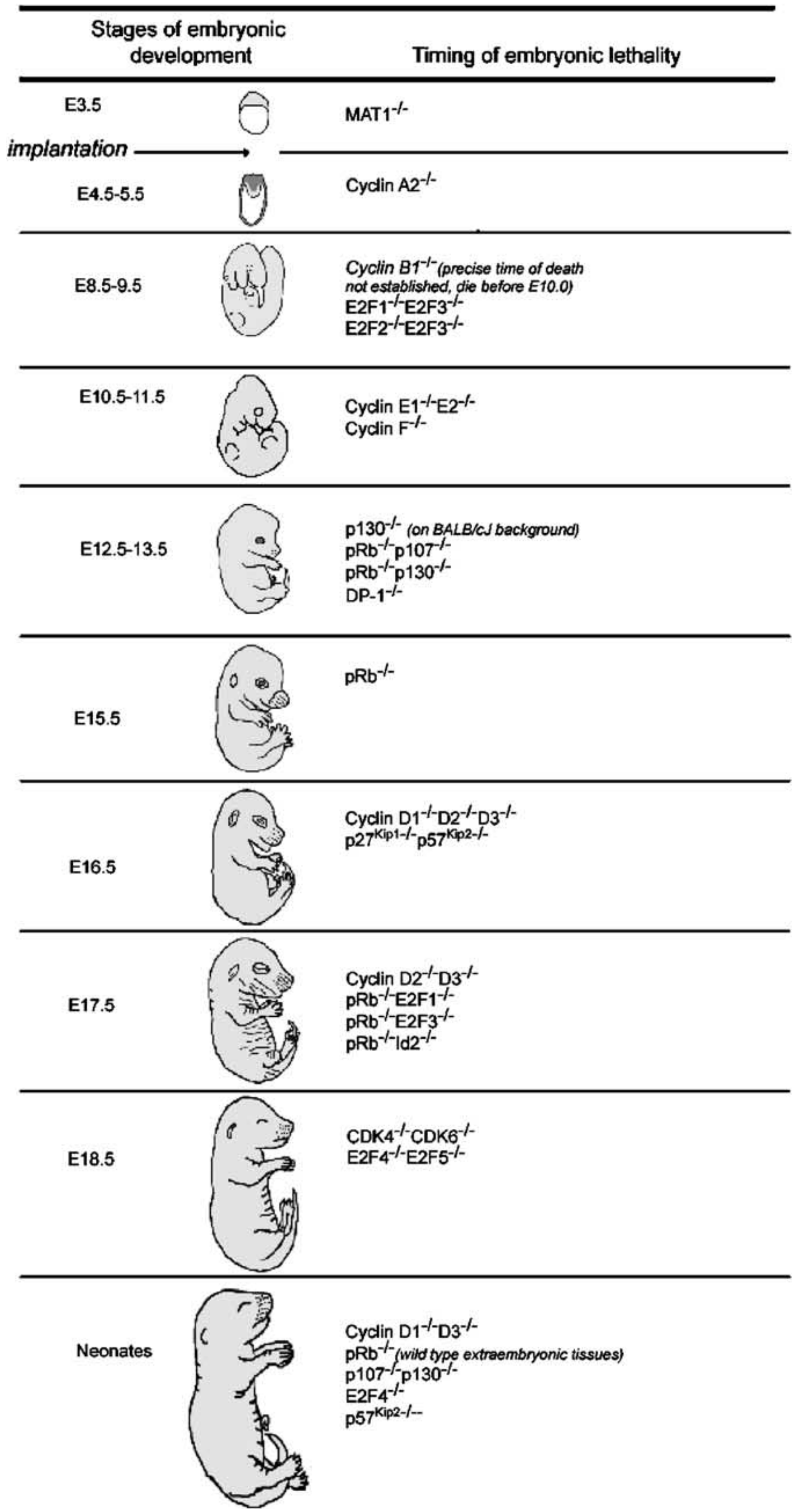

Figure 1 Timing of embryonic lethality associated with loss of one or more cell cycle regulators. Embryos are presented not to scale

Table 1 Developmental phenotypes associated with the loss of one or more cell cycle regulators

Genes disrupted	*Survival*	*Phenotypes*	*References*
Cyclin A1	Viable	Infertile males (small testes, spermatogenesis arrested at the first meiotic prophase, increased apoptosis)	Liu *et al.* (1998)
Cyclin A2	Embryonic lethal (E5.5)	Embryonic lethality shortly after implantation	Murphy *et al.* (1997)
Cyclin B1	Embryonic lethal (die before E10)	The cause and precise timing of embryonic lethality not determined	Brandeis *et al.* (1998)
Cyclin B2	Viable	No detectable abnormalities	Brandeis *et al.* (1998)
Cyclin D1	Viable	Reduced body size, neurological abnormalities, hypoplastic retinas, impaired proliferation of mammary gland epithelium during pregnancy	Fantl *et al.* (1995); Sicinski *et al.* (1995)
Cyclin D2	Viable	Infertile females (inability of ovarian granulosa cells to proliferate in response to FSH; oocyte development not affected), males fertile but have small testes and decreased sperm counts, impaired cerebellar and pancreatic β-cell development, impaired proliferation of B lymphocytes	Sicinski *et al.* (1996); Huard *et al.* (1999); Lam *et al.* (2000); Solvason *et al.* (2000); Georgia and Bhushan (2004); Kushner *et al.* (in press)
Cyclin D3	Viable	Hypoplastic thymuses (reduced expansion of immature T lymphocytes)	Sicinska *et al.* (2003)
Cyclins D1 and D2	Viable but die within 3 weeks after birth	Reduced body size, hypoplastic cerebella	Ciemerych *et al.* (2002)
Cyclins D1 and D3	Most die immediately after birth, a fraction can survive up to 2 months	Neonatal lethality due to aspiration with meconium (probably caused by neurological defects)	Ciemerych *et al.* (2002)
Cyclins D2 and D3	Embryonic lethal (E17.5–18.5)	Severe megaloblastic anemia	Ciemerych *et al.* (2002)
Cyclins D1–D3	Embryonic lethal (E16.5)	Severe megaloblastic anemia, multilineage hematopoietic failure, abnormal heart development	Kozar *et al.* (2004)
Cyclin E1	Viable	No detectable abnormalities	Geng *et al.* (2003); Parisi *et al.* (2003)
Cyclin E2	Viable	Decreased male fertility	Geng *et al.* (2003); Parisi *et al.* (2003)
Cyclins E1 and E2	Embryonic lethal (E11.5)	Abnormal placental development; failure of trophoblast giant cells to undergo normal endoreplication. Tetraploid complementation (development within wild-type extraembryonic tissues) prevented the embryonic lethality. Rescued embryos displayed cardiac abnormalities	Geng *et al.* (2003); Parisi *et al.* (2003)
Cyclin F	Embryonic lethal (E10.5)	Delayed embryonic development, abnormal development of the extraembryonic tissues (abnormal yolk sac and chorioallantoic placental development)	Tetzlaff *et al.* (2004)
CDK2	Viable	Reduced body size, females infertile due to abnormal oogenesis, males infertile due to the failure of spermatogenesis (defective chromosome pairing)	Berthet *et al.* (2003); Ortega *et al.* (2003)
CDK4	Viable	Reduced body size, pancreatic dysfunction resulting in diabetes (β-cell hypoplasia), male and female infertility	Rane *et al.* (1999); Tsutsui *et al.* (1999); Mettus and Rane, (2003)
CDK6	Viable	Hypoplasia of thymuses and spleens, reduced number of megakaryocytes and erythrocytes	Malumbres *et al.* (2004)
CDK4 and CDK6	Embryonic lethal (E14.5–E18.5)	Severe anemia, defects in maturation of different hematopoietic lineages	Malumbres *et al.* (2004)
pRb	Embryonic lethal (E12.0–E15.0)	Defects in erythroid, neuronal and lens development, severe abnormalities in placental development. Tetraploid complementation (development within wild-type extraembryonic tissues) resulted in completion of embryonic development	Clarke *et al.* (1992); Jacks *et al.* (1992); Lee *et al.* (1992); Wu *et al.* (2003)

p107	Viable	No obvious abnormalities in 129Sv/C57BL6 background. Growth retardation, myeloid hyperplasia in BALB/cJ background	Cobrinik *et al*. (1996); Lee *et al*. (1996); LeCouter *et al*. (1998a)
p130	Viable	No obvious abnormalities in 129Sv/C57BL6 background. Embryonic lethal (E11–E13) when generated in BALB/cJ background, with defects in neurogenesis and myogenesis	LeCouter *et al*. (1998b)
p107 and p130	Neonatally lethal	Generated in 129Sv/C57BL6 background. Abnormal chondrocyte proliferation, aberrant endochondral bone development	Cobrinik *et al*. (1996)
pRb and p107	Embryonic lethal (E11–E13)	Apoptosis in liver and central nervous system	Lee *et al*. (1996)
E2F1	Viable	Dysplasia of exocrine glands (salivary glands, pancreas), testicular atrophy, increased cellularity of thymus and lymph nodes	Field *et al*. (1996); Yamasaki *et al*. (1996)
E2F2	Viable	Lupus-like autoimmune disorder, increased proliferation of T-lymphocytes	Murga *et al*. (2001)
E2F3	Viable	Growth retardation, congestive heart failure. Generated in 129Sv background die *in utero*	Humbert *et al*. (2000b); Cloud *et al*. (2002); Saavedra *et al*. (2002)
E2F4	Neonatally lethal	Craniofacial abnormalities, defective development of intestinal tract and hematopoietic organs (spleen, thymus, bone marrow). Impaired erythrocyte maturation, anemia	Humbert *et al*. (2000a); Rempel *et al*. (2000)
E2F5	Viable	Abnormal cranial and extracranial development, overproduction of cerebrospinal fluid causing nonobstructive hydrocephalus and premature death at 6 weeks of age	Lindeman *et al*. (1998)
E2F6	Viable	Abnormalities of axial skeleton	Storre *et al*. (2002)
E2F1 and E2F2	Viable	Reduced lifespan, impaired pancreatic development	Li *et al*. (2003); Iglesia *et al*. (2004)
E2F1 and E2F3	Embryonic lethal (E9.5)	Growth retardation, cause of death not characterized	Wu *et al*. (2001); Cloud *et al*. (2002)
E2F2 and E2F3	Embryonic lethal (E9.5)	Cause of death not characterized	Wu *et al*. (2001)
E2F4 and E2F5	Embryonic lethal (E18.5)	Cause of death not characterized	Gaubatz *et al*. (2000)
DP-1	Embryonic lethal (E12.5)	Developmentally delayed embryos, defects in the extraembryonic tissues (abnormal yolk sac)	Kohn *et al*. (2003); Kohn *et al*. (2004)
Cdc25B	Viable	Females sterile due to failure to reinitiate meiosis during oocyte maturation	Lincoln *et al*. (2002)
Cdc25C	Viable	No obvious abnormalities	Chen *et al*. (2001)
Mat1	Embryonic lethal (E3.5)	Proliferation failure at the blastocyst stage	Rossi *et al*. (2001)
$p15^{INK4b}$	Viable	No obvious developmental abnormalities, except for extramedullary hematopoiesis, lymphoid hyperplasia in spleen and lymph nodes and the increased lymphocyte proliferation in response to mitogenic stimulation	Latres *et al*. (2000)
$p16^{INK4a}$	Viable	No obvious developmental abnormalities except for the increased lymphocyte proliferation in response to mitogenic stimulation	Krimpenfort *et al*. (2001); Sharpless *et al*. (2001)
$p18^{INK4c}$	Viable	Splenic and thymic hyperplasia in older mice	Franklin *et al*. (1998); Latres *et al*. (2000)
$p19^{INK4d}$	Viable	Male infertility due to testicular hypoplasia, progressive hearing loss	Zindy *et al*. (2001); Chen *et al*. (2003)
$p21^{Cip1}$	Viable	No obvious developmental abnormalities	Deng *et al*. (1995)
$p27^{Kip1}$	Viable	Increased body size, multiple organ hyperplasia, retinal dysplasia, female sterility due to defective cell cycle exit of corpus luteum cells	Fero *et al*. (1996); Kiyokawa *et al*. (1996); Nakayama *et al*. (1996)
$p21^{Cip1}$ and $p27^{Kip1}$	Viable	Phenotype similar to $p27^{kip-/-}$, with more pronounced hyperplasia of ovaries	Jirawatnotai *et al*. (2003)
$p57^{Kip2}$	Neonatally lethal	Abnormal proliferation in limbs, lens, and intestine. Cleft palate, organomegaly, defects in placental development	Yan *et al*. (1997); Zhang *et al*. (1997); Zhang *et al*. (1998)
$p27^{Kip1}$ and $p57^{Kip2}$	Embryonic lethal (E12-E16.5)	Proliferative disorders similar to observed in $p57^{kip2-/-}$ mice, more pronounced abnormalities in lens	Zhang *et al*. (1998)

redundant functions in development. Double mutants are characterized by abnormal chondrocyte proliferation that results in aberrant endochondral bone and shortened limbs (Cobrinik *et al*., 1996). Interestingly, pRb$^{-/-}$p107$^{-/-}$ or pRb$^{-/-}$p130$^{-/-}$ mutants displayed a similar phenotype, further supporting the notion that pRb family members can substitute for each other. These double mutants died earlier than pRb-deficient littermates (between days 11 and 13 of embryonic development) and showed pronounced apoptosis in the liver and within the CNS (Lee *et al*., 1996). It cannot be excluded, however, that the observed embryonic lethality is caused by enhanced placental defects characteristic for pRb-deficient embryos.

The consequences of the loss of all three pRb family members have been studied only using *in vitro* cultured cells. pRb$^{-/-}$p107$^{-/-}$p130$^{-/-}$ mouse embryonic stem (ES) cells displayed normal proliferation rates (Dannenberg *et al*., 2000; Sage *et al*., 2000), supporting the notion that these early embryonic cells do not require pocket proteins for proliferation (Iwamori *et al*., 2002). However, triple knockout ES cells had limited differentiation potential. Whereas wild-type ES cells can differentiate into variety of cell types including neuronal and muscle cells upon subcutaneous inoculation into nude mice, triple knockout ES cells were able to differentiate only into primitive neuronal cells (Dannenberg *et al*., 2000). *In vitro* cultured pRb$^{-/-}$p107$^{-/-}$p130$^{-/-}$ mouse embryonic fibroblasts were unable to arrest their proliferation in response to confluence, low serum or other inhibitory signals. Moreover, these cells showed increased proliferation rate, demonstrating that pRb family members do play an important role in cell cycle regulation of somatic cells (Dannenberg *et al*., 2000; Sage *et al*., 2000).

It is well established that the retinoblastoma protein functions as a tumor suppressor. Inactivation of *Rb* gene in humans leads to development of childhood eye tumors, retinoblastomas (Friend *et al*., 1986; Lee *et al*., 1987). pRb has also been shown to be responsible for the development of osteosarcomas, prostate and pituitary tumors, lymphomas and several other human malignancies (reviewed by Malumbres and Barbacid, 2001). Consistent with the tumor-suppressor function for pRb, heterozygous pRb$^{+/-}$ mice developed pituitary adenocarcinomas of the intermediate lobe, thyroid medullary carcinomas and neuroendocrine tumors (Jacks *et al*., 1992; Hu *et al*., 1994; Maandag *et al*., 1994; Williams *et al*., 1994a; Harrison *et al*., 1995; Nikitin *et al*., 1999). Appearance of these tumors is accompanied by the loss of the wild-type pRb allele (Hu *et al*., 1994; Nikitin and Lee, 1996). Notably, pRb$^{+/-}$ mice never develop retinoblastomas (Jacks *et al*., 1992).

The pRb$^{+/-}$ mouse model has been extensively used to investigate the relationship between pRb and other cell cycle regulators. It has been shown that the frequency of pituitary and thyroid tumors in pRb$^{+/-}$ mice can be reduced by the ablation of E2F1 (Yamasaki *et al*., 1998) or E2F4 (Lee *et al*., 2002). Conversely, ablation of a CDK inhibitor, p27^{Kip1}, accelerated tumorigenesis in pRb$^{+/-}$ mice. p27^{Kip1}-null mice, similarly to pRb$^{+/-}$ mice, develop intermediate lobe pituitary tumors (Fero *et al*., 1996; Kiyokawa *et al*., 1996; Nakayama *et al*., 1996). Loss of the remaining wild-type pRb allele in the pituitaries of pRb$^{+/-}$p27$^{kip1-/-}$ mice resulted in much earlier development of more aggressive pituitary tumors, thereby providing genetic evidence that pRb and p27^{Kip1} cooperate in the process of tumor suppression (Park *et al*., 1999).

As stated above, ablation of pRb is not sufficient to cause retinoblastoma in mice. Since in chimeric embryos, pRb$^{-/-}$ cells were shown to be eliminated from the retinas, it was postulated that cells lacking pRb undergo apoptosis and are hence not able to cause retinoblastomas (Maandag *et al*., 1994). This notion was supported by recent studies showing that lack of pRb causes failure to exit the cell cycle and results in apoptosis of certain retinal cells (MacPherson *et al*., 2004). However, it was shown that cells lacking both pRb and p107 can give rise to retinoblastomas (Robanus-Maandag *et al*., 1998). Likewise, retinal-specific ablation of pRb in the p107$^{-/-}$ or p130$^{-/-}$ background resulted in retinoblastomas (Chen *et al*., 2004; MacPherson *et al*., 2004).

E2Fs

The best-characterized targets of the ‘pocket proteins’ are members of E2F family of transcription factors (E2F1–E2F7). E2Fs heterodimerize with two partners, called DP1 and DP2 (Zhang and Chellappan, 1995; reviewed by Trimarchi and Lees, 2002). E2F factors can be divided into two groups that differently influence transcription, namely activatory E2Fs (E2F1, E2F2 and E2F3a) and repressive E2Fs (E2F4, E2F5 and E2F6) (reviewed by Blais and Dynlacht, 2004); the recently characterized E2F7 acts independently of DP partners and suppresses gene expression (de Bruin *et al*., 2003; Di Stefano *et al*., 2003; Logan *et al*., 2004). Among the E2F targets are genes encoding cell cycle regulators, such as cyclins E and A, p107, E2F1, as well as proteins involved in DNA replication (see Dimova and Dyson, *Oncogene*, this issue; also reviewed by Muller *et al*., 2001; Bracken *et al*., 2004).

Analyses of E2F1, E2F3, E2F5 and E2F6 expression revealed that during preimplantation development, their levels fluctuate and that they all become steadily expressed at the time of implantation (Palena *et al*., 2000). During postimplantation development, each of six E2Fs (E2F1–6) is expressed in a tissue-specific manner (Tevosian *et al*., 1996; Dagnino *et al*., 1997a, b; Kusek *et al*., 2000).

Mice lacking E2F1 were viable (Field *et al*., 1996; Yamasaki *et al*., 1996) and displayed dysplasia of exocrine glands (salivary glands, pancreas) and testicular atrophy (Yamasaki *et al*., 1996). The loss of E2F1 also resulted in increased size and cellularity of thymuses and lymph nodes. This phenotype resulted from abnormal expansion of mature CD4^{+} or CD8^{+} single positive thymocytes and peripheral T-cells due to a defect in T-lymphocyte apoptosis (Field *et al*., 1996;

Garcia *et al*., 2000). Moreover, in older animals, the proliferation of immature thymocytes was dramatically increased, suggesting that E2F1 also plays a role as a suppressor of cellular proliferation (Field *et al*., 1996). Very unexpectedly, E2F1$^{+/-}$ and E2F1$^{-/-}$ mice developed a broad spectrum of tumors, such as reproductive tract sarcomas, lung tumors and lymphomas (Yamasaki *et al*., 1996). This surprising finding demonstrated an unexpected role for E2F1 role in tumor suppression.

Ablation of a single allele of E2F3 in E2F1$^{-/-}$ background caused even more pronounced growth retardation and testicular atrophy than those observed in E2F1-null mice (Cloud *et al*., 2002). Importantly, E2F3 deficiency did not increase tumor incidence in E2F1$^{+/-}$ mice, suggesting that tumor suppression is a specific property of E2F1 but not E2F3 (Cloud *et al*., 2002).

Mice lacking E2F2 displayed increased proliferation of T lymphocytes upon TCR stimulation. Moreover, mutant mice frequently developed lupus-like autoimmune disorder (Murga *et al*., 2001). Double knockout E2F1$^{-/-}$E2F2$^{-/-}$ mice were viable, but they displayed a reduced lifespan. These animals developed diabetes and exocrine pancreatic insufficiency, indicating that E2F1 and E2F2 are required for pancreatic development and function (Li *et al*., 2003; Iglesias *et al*., 2004).

E2F3-deficient mice were engineered to lack both E2F3a and E2F3b transcription factors, two splice variants encoded by the same locus (Humbert *et al*., 2000b). When maintained in a pure 129/Sv genetic background, mutant animals died *in utero*. In contrast, a fraction of animals generated in a mixed genetic background (C57BL/6 × 129/Sv) was viable, suggesting the existence of a strain-specific modifier. These E2F3-null mice in C57BL/6 × 129/Sv background displayed growth retardation, and started to die at 7 months of age due to congestive heart failure (Cloud *et al*., 2002). Importantly, ablation of one allele of pRb in E2F3$^{-/-}$ mice rescued these heart abnormalities, most likely by increasing the levels of free E2Fs (Cloud *et al*., 2002). Combined ablation of E2F1 and E2F3, or E2F2 and E2F3 resulted in embryonic lethality around day 9.5 of pregnancy, revealing overlapping functions for these E2Fs in development (Cloud *et al*., 2002).

The phenotype of triple knockout E2F1$^{-/-}$E2F2$^{-/-}$ E2F3$^{-/-}$ mice has not been described. However, cells deficient in E2F1, E2F2 and E2F3 were generated by Wu *et al*. (2001). Ablation of all the three E2F family members completely blocked the proliferation of mouse embryonic fibroblasts, indicating an essential requirement for these proteins in cell cycle progression.

E2F4 and E2F5 are primarily involved in the repression of E2F-responsive genes through the recruitment of histone deacetylases associated with p107 and p130 (reviewed by Grana *et al*., 1998). Ablation of the *E2F4* gene caused neonatal lethality, craniofacial defects, anemia and growth deficiency of mutant mice (Humbert *et al*., 2000a; Rempel *et al*., 2000). Histological analyses of E2F4$^{-/-}$ embryos also revealed defects in the intestinal tract and in the hematopoietic tissues (spleen, thymus, bone marrow) (Dagnino *et al*., 1997b; Rempel *et al*., 2000). Within the hematopoietic lineages, E2F4-null animals displayed decreased proportion of mature cells with concomitant increased proportion of immature cells such as reticulocytes, nucleated erythrocytes, granulocyte precursors and immature T-cells in thymuses (Humbert *et al*., 2000a; Rempel *et al*., 2000). *In vitro* cultured E2F4-null fibroblasts showed a dramatic decrease in the levels of total cellular E2Fs. Surprisingly, the cell cycle progression and expression of E2F-responsive genes remained unaffected in these cells (Humbert *et al*., 2000a).

E2F5-deficient mice underwent normal embryonic development. The animals suffered from nonobstructive hydrocephalus that was most likely caused by the overproduction of cerebrospinal fluid by chorioid plexus epithelium. This led to premature death at the age of 6 weeks (Lindeman *et al*., 1998). Since combined inactivation of E2F4 and E2F5 resulted in late embryonic lethality, the two E2Fs may play overlapping functions in development (Gaubatz *et al*., 2000). Interestingly, mouse embryonic fibroblasts lacking E2F4 and E2F5 (but not single knockout E2F4$^{-/-}$ or E2F5$^{-/-}$ cells) are resistant to cell cycle arrest by p16^{INK4a} (Gaubatz *et al*., 2000).

E2F6 – retinoblastoma-independent transcriptional repressor – is ubiquitously expressed in a variety of tissues with higher expression in the heart and skeletal muscle (Kherrouche *et al*., 2001; Dahme *et al*., 2002). Mice lacking E2F6 were viable and healthy (Storre *et al*., 2002). However, mutant animals displayed abnormalities in axial skeleton that resembled the defects observed in the mice lacking Bmi-1 polycomb protein known to associate with E2F6 (van der Lugt *et al*., 1994; Trimarchi *et al*., 2001). These observations raise a possibility that during the development of certain tissues, E2F6 might serve to recruit polycomb factors to their target promoters (van der Lugt *et al*., 1994; Storre *et al*., 2002). However, other defects characteristic for Bmi-1 deficiency, such as defective hematopoiesis and cerebellar abnormalities, were not observed in E2F6-null animals (van der Lugt *et al*., 1994; Storre *et al*., 2002; Park *et al*., 2003; Leung *et al*., 2004).

In summary, analyses of E2F-deficient mouse strains delineated the distinct functions of the E2F family members in development and in cell cycle progression. Moreover, these studies unexpectedly revealed a function of E2F-1 as a tumor suppressor.

DP-1

The function of E2F transcription factors requires heterodimerization with members of the DP family, DP-1 or DP-2 (Helin *et al*., 1993; Wu *et al*., 1995; Zhang and Chellappan, 1995). During mouse development, high DP-1 expression is observed in the embryo proper as well as within the extraembryonic tissues. The levels of DP-1 decrease in adult tissues, where DP-1 remains ubiquitously expressed at lower levels (Wu *et al*., 1995;

Gopalkrishnan *et al.*, 1996; Tevosian *et al.*, 1996; Kohn *et al.*, 2003). Also, DP-2 is detectable both within the embryo proper and in placentas. In adult tissues, however, DP-2 transcripts are present in a restricted set of tissues including the heart, skeletal muscle and kidney, but not in the brain and lungs (Wu *et al.*, 1995; Rogers *et al.*, 1996).

Inactivation of the *DP-1* gene resulted in embryonic lethality by day 12.5 of pregnancy. At earlier stages of development (embryonic day 9.5–10.5), mutant embryos were developing within unusually small yolk sacs and displayed a severe developmental delay (Kohn *et al.*, 2003). Detailed analyses revealed that the development of extraembryonic tissues was significantly affected by DP-1 deficiency. Specifically, ablation of DP-1 perturbed development of the ectoplacental cone. In addition, trophoblast giant cells displayed DNA replication failure. All these extraembryonic abnormalities might have led to the lethality observed in DP-1-null conceptuses (Kohn *et al.*, 2003). This possibility was further tested by injecting DP-1$^{-/-}$ ES cells into wild-type blastocysts and by creating chimeric embryos. DP-1-null cells readily contributed to the embryo proper of DP-1$^{-/-}$ $\times$ wild-type chimeras, consistent with the notion that DP-1 is dispensable for the proliferation of cells within the embryo (Kohn *et al.*, 2004). Collectively, these studies suggest that DP-1, like the retinoblastoma protein, plays an important role in the proliferation of placental, trophectoderm-derived lineages.

Cyclin E-CDK2

Two E-type cyclins, cyclin E1 and E2, have been described (Sherr and Roberts, 2004). E-type cyclins represent critical cell cycle targets of the E2F/DP complexes (see Dimova and Dyson, *Oncogene*, this issue). Once synthesized, E-cyclins bind CDK2 and phosphorylate several cellular targets such as the retinoblastoma protein (Lundberg and Weinberg, 1998; Harbour *et al.*, 1999), p27^{Kip1} (Sheaff *et al.*, 1997; Vlach *et al.*, 1997), Cdc25A (Hoffmann *et al.*, 1994), proteins involved in centrosome duplication such as nucleophosmin and CP110 (Okuda *et al.*, 2000; Chen *et al.*, 2002), p220^{NPAT} required for histone biosynthesis (Ma *et al.*, 2000), E2F5 (Morris *et al.*, 2000), p300/CBP (Perkins *et al.*, 1997; Ait-Si-Ali *et al.*, 1998; Felzien *et al.*, 1999), as well as proteins involved in DNA replication (Krude *et al.*, 1997; Arata *et al.*, 2000; Zou and Stillman, 2000; Furstenthal *et al.*, 2001). It was assumed that cyclins E1 and E2, together with their catalytic partner, CDK2, represented the essential components of the core cell cycle machinery. This notion was based on results of *in vitro* studies in which cell cycle progression was blocked by injections of antibodies against cyclin E or CDK2 (Tsai *et al.*, 1993; Ohtsubo *et al.*, 1995). Moreover, ectopic expression of the dominant-negative CDK2 blocked cell proliferation (van den Heuvel and Harlow, 1993). The first evidence that cell cycling can proceed in the absence of CDK2 came from the work of Tetsu and McCormick (2003), who demonstrated that inhibition of CDK2 activity or depletion of CDK2 by small interfering RNA did not block the proliferation of certain tumor cells. In a stunning development, Ortega *et al.* (2003) and Berthet *et al.* (2003) generated mice lacking CDK2 and observed that this protein was not essential for mouse viability. CDK2-null mice developed normally except for male and female infertility. Detailed analyses of the mutant animals revealed an unexpected role role for CDK2 in chromosome pairing the during the meiotic prophase (Berthet *et al.*, 2003; Ortega *et al.*, 2003).

In a parallel set of experiments, Geng *et al.* (2003) and Parisi *et al.* (2003) generated mice lacking E-type cyclins. Mice deficient in cyclin E1 or E2 were viable and developed normally, except for the testicular abnormalities seen in cyclin E2$^{-/-}$ males (Geng *et al.*, 2003; Parisi *et al.*, 2003). However, combined ablation of cyclins E1 and E2 led to lethality at embryonic day 11.5, revealing that the two proteins perform overlapping functions in mouse development. Unexpectedly, Geng *et al.* (2003) found that the embryonic lethality seen in cyclin E-null mice could be rescued by providing mutant embryos with wild-type extraembryonic tissues, using a technique of 'tetraploid blastocyst complementation' ('rescued' animals died at birth due to lung abnormalities, a phenotype not caused by cyclin E-deficiency but rather being a consequence of the technique used to generate the embryos (Tanaka *et al.*, 1997; Eggan *et al.*, 2001). Analyses of cyclin E-deficient placentas revealed that trophoblast giant cells, a cell type that plays critical role in placental development (Cross, 2000), failed to undergo normal endocycles in the absence of cyclin E. Likewise, the endoreplication in megakaryocytes was crippled in the absence of E-cyclins (Geng *et al.*, 2003; Parisi *et al.*, 2003).

Collectively, these findings demonstrated that the E-cyclins and their partner, CDK2, are dispensable for development of the embryo proper. Surprisingly, however, the phenotypes associated with cyclin E deficiency were more pronounced that those associated with the loss of CDK2. This was true both for the abnormalities observed in mice as well as in *in vitro* cultured CDK2$^{-/-}$ and cyclin E1$^{-/-}$E2$^{-/-}$ mouse embryo fibroblasts (the latter, but not former, displayed defect in cell cycle re-entry from the quiescent, G0 state, and they were resistant to the oncogenic transformation) (Geng *et al.*, 2003). One explanation for these discrepancies is that in the tissues of CDK2$^{-/-}$ mice, E-type cyclins activate another kinase, which can substitute for CDK2 function. An alternative possibility is the presence of kinase-independent functions for the E-type cyclins. Consistent with this possibility, Ortega *et al.* (2003) and Berthet *et al.* (2003) failed to detect any kinase activity associated with cyclin E in CDK2-null cells.

The placental phenotype seen in cyclin E-deficient mice is intriguing, since the same organ was affected in mice lacking the retinoblastoma protein (Wu *et al.*, 2003), DP-1 (Kohn *et al.*, 2003), as well as an F-box protein cyclin F (Tetzlaff *et al.*, 2004). It seems that the development of the extraembryonic tissues depends on these 'essential' cell cycle regulators, while the cells of the embryo proper are endowed with alternative pathways that allow them to proliferate in the absence of these proteins. These findings underscore the importance of studying the core cell cycle machinery in a cell type specific context.

Cyclin C–CDK3

Cyclin C was originally cloned in a screen for human and *Drosophila* cDNAs that were able to rescue lethality of *Saccharomyces cerevisiae* mutants lacking three CLN genes (Leopold and O'Farrell, 1991; Lew *et al*., 1991). Subsequently, the role for cyclin C in cell cycle control has been questioned. Instead, cyclin C, together with CDK8, was shown to represent a component of the RNA polymerase II holoenzyme complex. In this setting, cyclin C–CDK8 functions to phosphorylate the carboxy-terminal domain (CTD) of the largest subunit of RNA polymerase II (Tassan *et al*., 1995; Leclerc *et al*., 1996; Rickert *et al*., 1996). However, recent evidence indicates that cyclin C, acting together with another catalytic partner, CDK3, plays an important role in cell cycle re-entry from the quiescent state. Cyclin C–CDK3 complexes were shown to play a role in this process by contributing to pRb phosphorylation (Ren and Rollins, 2004). However, most strains of laboratory mice contain a premature chain termination mutation within the CDK3 locus, and are hence CDK3-null (Ye *et al*., 2001). For this reason, the cell cycle role of cyclin C in mouse tissues requires further studies.

Major M-phase regulators

Cyclins A and B, together with their kinase partner CDK1 (CDC2), represent the major regulators of the G2- and M-phase progression. Cyclin A also activates CDK2, and these various cyclin A–CDK complexes play roles in the S phase as well as in G2→M progression (Girard *et al*., 1991; Pagano *et al*., 1992). Cyclin A is synthesized at the onset of the S phase and was shown to phosphorylate proteins involved in DNA replication, such as CDC6 (Petersen *et al*., 1999; Coverley *et al*., 2000). During the G2→M transition, cyclin A–CDK1 activity is required for the initiation of prophase (Furuno *et al*., 1999). At the mitotic exit, cyclin A degradation precedes degradation of cyclin B and was shown to be required for chromosomal alignment and progression to anaphase (Draetta *et al*., 1989; Minshull *et al*., 1990; den Elzen and Pines, 2001).

Two A-type cyclins, A1 and A2, have been described in mammalian cells (Ravnik and Wolgemuth, 1996; Sweeney *et al*., 1996). Cyclin A1 protein levels are high in unfertilized oocytes; cyclin A1 is rapidly replaced by cyclin A2 as soon as embryonic development begins (Fuchimoto *et al*., 2001). These observations suggest that cyclin A1 may control meiosis, while cyclin A2 is responsible for directing mitotic divisions. Maternal mRNA encoding cyclin A2 is degraded at 2→4-cell transition (Winston *et al*., 2000). Hence, starting from a four-cell stage embryo, cyclin A2 is synthesized from the newly transcribed, embryonic mRNA. During postnatal development and in the adult mice, all proliferating compartments express cyclin A2. In contrast, expression of cyclin A1 is restricted mainly to the male and female germ cells, and to the brain (Ravnik and Wolgemuth, 1996; Sweeney *et al*., 1996; Yang *et al*., 1997).

Mice lacking cyclins A1 or A2 have been generated. Cyclin A1-null animals were viable with cyclin A1$^{-/-}$ females displaying normal fertility. In contrast, cyclin A1 deficiency in males resulted in an arrest of spermatogenesis before the first meiotic division, leading to increased germ cell apoptosis and infertility. In the mutant spermatocytes, CDK1 activation did not occur at the end of the meiotic prophase, and hence the first meiotic division could not be triggered (Liu *et al*., 1998, 2000).

In contrast to these narrow phenotypes of cyclin A1 deficiency, disruption of cyclin A2 led to an early embryonic lethality. Cyclin A2$^{-/-}$ embryos successfully underwent early development and reached blastocyst stage, but then died soon after the implantation (Murphy *et al*., 1997). Since maternal cyclin A2 transcripts and protein are degraded before an embryo reaches a four-cell stage (Winston *et al*., 2000), and yet cyclin A2-null embryos continued to proliferate for a few more days (Murphy *et al*., 1997), these observations indicate that cyclin A2 is dispensable for the early preimplantation development. It is possible that at this stage of development other proteins may replace the functions of cyclin A2. For instance, it was speculated that cyclin B3, which shares the homology with the A-type cyclins, may replace cyclin A2's functions (Winston *et al*., 2000). In this respect, progression of early embryonic cell cycles differs from that of somatic cells, since in cultured mammalian cells cyclin A2 was shown to play a critical, nonredundant role in both the S- and M-phase progression (Pagano *et al*., 1992; Resnitzky *et al*., 1995; Furuno *et al*., 1999).

B-type cyclins represent the important controllers of the M-phase progression. Three mammalian B-type cyclins has been described: B1 (Pines and Hunter, 1989; Chapman and Wolgemuth, 1992), B2 (Chapman and Wolgemuth, 1993) and B3 (Gallant and Nigg, 1994; Lozano *et al*., 2002; Nguyen *et al*., 2002). Cyclins B1 and B2 associate with CDK1 (Draetta *et al*., 1989; Pines and Hunter, 1989), while mammalian cyclin B3 was shown to interact with CDK2 but not with CDK1 (Nguyen *et al*., 2002). Cyclin B–CDK1 complexes operate during the M phase and regulate series of crucial changes accompanying cell division (reviewed by Obaya and Sedivy, 2002).

Cyclins B1 and B2 are expressed in the majority of proliferating cells, but they localize into distinct cellular compartments. Thus, cyclin B1 is a microtubule-associated protein, while cyclin B2 associates with the intracellular membranes (Ookata *et al*., 1993; Jackman *et al*., 1995). Moreover, cyclin B1, but not B2, translocates into the nucleus at the end of the G2 phase, suggesting that the two proteins play different functions during cell cycle progression (Pines and Hunter, 1991; Hagting *et al*., 1998; Toyoshima *et al*., 1998; Yang *et al*., 1998). Indeed, it has been shown that nuclear cyclin B1–CDK1 complexes are responsible for nuclear envelope breakdown, chromosome condensation and mitotic spindle assembly, while cytoplasmic cyclin B2–CDK1 complexes play a role in the mitotic reorganization of the Golgi apparatus (Draviam *et al*., 2001).

Cyclin B1 is an essential gene. Its deletion caused the embryonic lethality before day 10 of development (Brandeis *et al*., 1998). However, the reason of embryonic lethality in cyclin B1-deficient mice has not been established yet. In contrast, cyclin B2-deficient mice developed normally and did not display any obvious abnormalities with both females and males being fertile (Brandeis *et al*., 1998).

The third member of the mammalian B-type cyclin family – cyclin B3 – shares characteristics of both A- and B-type cyclins (Nieduszynski *et al*., 2002). Like cyclin A, cyclin B3 is localized exclusively in the cell nucleus (Gallant and Nigg, 1994) and it interacts with CDK2 (Nguyen *et al*., 2002). In contrast to cyclins B1 and B2, which are ubiquitously expressed, the expression of cyclin B3 is limited to the testes and to fetal, but not adult ovaries (Nguyen *et al*., 2002). The *in vivo* functions of cyclin B3 in mouse development have not been addressed so far, as no cyclin B3-null mice have been generated.

CDK1 represents the major kinase subunit that facilitates the action of cyclins during the G2→M phase progression. For this reason, one could expect that disruption of the *CDK1* gene would lead to an early embryonic lethality. However, this has not been tested, as mouse strain deficient in CDK1 has not been reported.

Cdc25 family members

Cyclin–CDK complexes are held in an inactive state by inhibitory phosphorylations of the CDK molecules (Russell and Nurse, 1987; Lundgren *et al*., 1991; McGowan and Russell, 1993). These inhibitory phosphorylations are removed by the action of Cdc25 phosphatases (reviewed by Donzelli and Draetta, 2003). Hence, the Cdc25 proteins catalyse a key step of cyclin–CDK activation.

In mammals, Cdc25 phosphatases are encoded by three related genes – *Cdc25A*, *Cdc25B* and *Cdc25C*. Cdc25A is responsible for the regulation of both G1→S and G2→M-phase transitions (Hoffmann *et al*., 1994; Jinno *et al*., 1994; Mailand *et al*., 2002), while Cdc25B and Cdc25C control the G2→M transition (Gabrielli *et al*., 1996; Lammer *et al*., 1998; Karlsson *et al*., 1999). Acting in the late G1 phase, Cdc25 dephosphorylates cyclin–CDK2 complexes (Hoffmann *et al*., 1994), while at the G2→M phase transition it activates cyclin–CDK1 holoenzyme (Lammer *et al*., 1998; Mailand *et al*., 2002).

During early embryo development, Cdc25A transcripts are absent until the the blastocyst stage. Later, Cdc25A expression is ubiquitous and virtually unchanged during the whole period of embryonic development. In adult tissues, the highest Cdc25A levels were observed in female and male germlines (Kakizuka *et al*., 1992; Wickramasinghe *et al*., 1995; Wu and Wolgemuth, 1995). Importantly, the *in vivo* finctions of Cdc25A have not been tested yet by generating Cdc25A-deficient mice.

In contrast to Cdc25A, Cdc25B is expressed during early embryo development. Cdc25B is detectable at a one-cell stage, its levels transiently decrease at a two-cell stage and raise again at a four-cell stage, concomitant with embryonic genome activation (Wickramasinghe *et al*., 1995). During postimplantation development, Cdc25B peaks at day 13.5 and then decreases at day 17.5, when it is expressed predominantly in the liver and neuronal tissues (Kakizuka *et al*., 1992; Wickramasinghe *et al*., 1995). Its expression in female and male germ cells is low (Wickramasinghe *et al*., 1995). Surprisingly, Cdc25B-null animals were viable and displayed no obvious phenotypes except for the female sterility. In the mutant females, oocyte maturation was arrested at the meiotic prophase due to low levels of cyclin B-CDK1; this activity is essential for the reinitiation of meiosis (Lincoln *et al*., 2002). In contrast, Cdc25B$^{-/-}$ embryonic fibroblasts proliferated normally, pointing to the crucial role of Cdc25B in meiotic, but not mitotic, divisions (Lincoln *et al*., 2002).

Cdc25C is expressed in rapidly proliferating cellular compartments of the developing mouse embryo (Wu and Wolgemuth, 1995). In the adult tissues, the highest levels of Cdc25C transcripts were detected in the ovaries and testes (Wu and Wolgemuth, 1995; Mitra and Schultz, 1996). Mice lacking Cdc25C were viable and displayed no obvious phenotypes including normal fertility (Chen *et al*., 2001). Proliferation rates of Cdc25C$^{-/-}$ fibroblasts did not differ from those seen in wild-type controls (Chen *et al*., 2001). Moreover, phosphorylation status of the Cdc25C target – CDK1 – was comparable between mutant and control cells, suggesting that other family members may compensate for the lack of Cdc25C (Chen *et al*., 2001).

Collectively, these findings suggest that the three Cdc25 proteins play overlapping functions during mouse embryo development. Hence, generation and analyses of mice lacking two or even three Cdc25 family members will be necessary to address the functions played by these proteins.

CAK

Another level of control of cyclin–CDK activity is provided by the CDK-activating kinase (CAK). This enzyme, composed of cyclin H, a kinase subunit CDK7, and an assembly protein Mat1, catalyses activating phosphorylation of cyclin-CDK complexes at a threonine residue of the CDK. In addition, cyclinH/CDK7/Mat1 complexes were shown to constitute a subunit of basal transcription factor TFIIH (reviewed by Kaldis, 1999).

In vivo functions of the CAK complex have been tested by genetic ablation of the *Mat1* gene. Mat1-deficient embryos did not display any proliferative defects at preimplantation stages when maternally inherited stocks of Mat1 protein or/and mRNA were still available. However, depletion of maternal Mat1 led to embryonic death occurring soon after the implantation (Rossi *et al.*, 2001). When cultured *in vitro*, inner cell masses of Mat1$^{-/-}$ blastocysts were unable to proliferate. In contrast, mutant trophoblast cells proliferated *in vitro*, but failed to undergo normal cycles of endoreduplication (Rossi *et al.*, 2001). One of the targets of cyclin H/CDK7/Mat1 kinase are cyclin E-CDK2 complexes. While CDK2-deficient embryos developed successfully without any visible abnormalities within the trophoblast lineage (Berthet *et al.*, 2003; Ortega *et al.*, 2003), ablation of the E-type cyclins resulted in the failure of trophoblast cells to undergo normal endoreplication (Geng *et al.*, 2003; Parisi *et al.*, 2003). Collectively, these observations raise a possibility that complexes of E-cyclins with other CDKs might represent targets for cyclin H/CDK7/Mat1 kinase in trophoblast cells.

In addition to these cell cycle defects, ablation of Mat1 also caused defects in phosphorylation of the C-terminal domain of RNA polymerase II (Rossi *et al.*, 2001). More recently, conditional ablation of Mat1 in male germ-cell lineage resulted in death of mitotically dividing spermatogonia, but it did not affect nondividing postmitotic germ cells (Korsisaari *et al.*, 2002). Consistent with these observations are the analyses of Schwann cells with conditional ablation of the *Mat1* gene, which revealed that postmitotic myelinating cells were able to attain a mature myelinated phenotype (Korsisaari *et al.*, 2002). Collectively, these studies highlight the crucial role of Mat1 in the control of cell proliferation.

INK and Cip/Kip families of CDK inhibitors

A very important role in the control of cell cycle progression is attributed to inhibitors of cyclin-dependent kinases. These inhibitors can be divided into two groups: INK and Kip/Cip families. The INK family encodes inhibitors of cyclin D-CDK4/6 complexes – p16^{INK4a}, p15^{INK4b}, p18^{INK4c} and p19^{INK4d} (Serrano *et al.*, 1993; Hannon and Beach, 1994; Chan *et al.*, 1995; Hirai *et al.*, 1995; Quelle *et al.*, 1995a; Guan *et al.*, 1996). The Kip/Cip family consists of inhibitors of cyclin/CDK complexes – p21^{Cip1}, p27^{Kip1} and p57^{Kip2} (Harper *et al.*, 1993; Xiong *et al.*, 1993; Polyak *et al.*, 1994; Lee *et al.*, 1995).

Analyses of the expression pattern of INK family members revealed that p16^{INK4a} and p15^{INK4b} are not expressed during embryonic development. p15^{INK4b} is first detectable in the organs isolated from 4-weeks-old animals, while p16^{INK4a} is expressed in nearly all tissues by 15 months of life (Zindy *et al.*, 1997a). A drastic upregulation of p16^{INK4a} was observed in the spleens and lungs of aging mice (Zindy *et al.*, 1997a). In contrast, p18^{INK4c} and p19^{INK4d} are expressed during embryonic development, as well as at later stages of animals' lives (Zindy *et al.*, 1997a, b). In adult animals, p18^{INK4c} transcripts are detectable in a majority of tissues except brain, spleen and liver, while p19^{INK4d} is expressed ubiquitously (Hirai *et al.*, 1995; Zindy *et al.*, 1997a, b).

Mice lacking p16^{INK4a} have been generated (Serrano *et al.*, 1996). However, at that time, it was unknown that the INK4a locus encoded two different polypeptides: p16^{INK4a} and p19^{ARF} (p14^{ARF} in humans) (Quelle *et al.*, 1995b). Hence, these INK4-deficient mice lacked both p16^{INK4a} and p19^{ARF}. Combined ablation of p16^{INK4a} and p19^{ARF} did not result in any developmental abnormalities; however, mutant mice developed a variety of tumors such as lymphomas and sarcomas (Serrano *et al.*, 1996). Subsequently, mice lacking only p16^{INK4a} (but expressing intact p19^{ARF}) were generated (Krimpenfort *et al.*, 2001; Sharpless *et al.*, 2001). Again, these mice displayed normal development with the exception of thymic hyperplasia, but they were characterized by a high rate of spontaneous and carcinogen-induced tumor occurrence (Krimpenfort *et al.*, 2001; Sharpless *et al.*, 2001, 2004). These analyses revealed that p16^{INK4a} acts as a tumor suppressor.

Mice lacking another member of the INK family, p15^{INK4b}, were also viable. However, mutant animals displayed extramedullary hematopoiesis and lymphoid hyperplasia in spleen and lymph nodes (Latres *et al.*, 2000). Lymphocytes derived from p15^{INK4b}-null mice, similarly to p16^{INK4a}-deficient lymphocytes, showed by increased proliferation rates in response to mitogenic stimulation (Latres *et al.*, 2000; Sharpless *et al.*, 2001). Moreover, p15^{INK4b}-deficient mice displayed low incidence of tumor development (hemangiocarcinomas) (Latres *et al.*, 2000).

Ablation of p18^{INK4c} also did not affect development (Franklin *et al.*, 1998; Latres *et al.*, 2000). However, with aging, p18^{INK4c}-null mice became larger than the control animals and displayed organomegaly with particularly disproportionate enlargement of spleens and thymuses (Franklin *et al.*, 1998). Deregulation of epithelial proliferation in p18^{INK4c}-deficient mice led to the formation of cysts in the cortical region of kidneys and in mammary gland epithelium (Franklin *et al.*, 1998; Latres *et al.*, 2000). Moreover, p18$^{INK4c-/-}$ mice, similarly to pRb$^{+/-}$ and p27$^{Kip1-/-}$ animals, developed pituitary tumors with high penetrance (Franklin *et al.*, 1998; Latres *et al.*, 2000). Ablation of both p18^{INK4c} and p27$^{Kip1-/-}$ resulted in drastic acceleration of pituitary tumor development, suggesting that the two proteins operate in different pathways (Franklin *et al.*, 2000). On the other hand, double knockout p18$^{INK4c-/-}$ p15$^{INK4b-/-}$ mice displayed the sum of phenotypes seen in mice lacking p18^{INK4c} or p15^{INK4b} alone, but showed only few additional phenotypes (enlarged testes, hyperplastic Langerhans islets, presence of cysts in these organs) (Latres *et al.*, 2000).

Mice lacking p19^{INK4d} were viable, but they displayed testicular hypoplasia and male infertility (Zindy *et al.*, 2000, 2001). Moreover, p19^{INK4d}-null animals have progressive hearing loss due to disruption of the maintenance of the postmitotic state of sensory hair

Table 2 Tumor incidence or tumor resistance associated with the loss of one or more cell cycle regulators

Mouse genotype		*Tumors developing in mutant animals*	*References*
Increasing tumor incidence	*Causing tumor resistance*		
	Cyclin D1$^{-/-}$	Resistant to breast cancers driven by activated Ras and c-ErbB-2	Yu *et al*. (2001)
		Reduced susceptibility to skin papillomas	Robles *et al*. (1998)
		Reduced susceptibility to intestinal polips caused by Apc^{Min} deficiency	Hulit *et al*. (2004)
	Cyclin D2$^{-/-}$	Reduced susceptibility to gonadal tumors	Burns *et al*. (2003)
	Cyclin D3$^{-/-}$	Resistant to Notch-driven leukemias. Delayed incidence of thymomas caused by p56^{LCK}	Sicinska *et al*. (2003)
	CDK4$^{-/-}$	Reduced susceptibility to skin papillomas and Myc-induced tumors of the oral mucosa	Rodriguez-Puebla *et al*. (2002); Miliani De Marval *et al*. (2004)
		CDK4 insensitive to INK4	
		Multiple tumor development including hemangiosarcomas, pituitary tumors, pancreatic endocrine tumors, melanomas	Sotillo *et al*. (2001a, b)
pRb$^{+/-}$		Pituitary adenocarcinomas of intermediate lobe, thyroid medullary carcinomas, neuroendocrine tumors	Jacks *et al*. (1992); Hu *et al*. (1994); Harrison *et al*. (1995); Nikitin *et al*. (1999)
pRb$^{-/-}$ and p107$^{-/-}$		Conditional ablation of pRb within retinas of p107$^{-/-}$ mice resulted in retinoblastomas	Chen *et al*. (2004); MacPherson *et al*. (2004)
		Retinoblastomas in pRb$^{-/-}$p107$^{-/-}$ chimeras	Robanus-Maandag *et al*. (1998)
pRb$^{-/-}$ and p130$^{-/-}$		Conditional ablation of pRb within retinas of p130$^{-/-}$ mice resulted in retinoblastomas	MacPherson *et al*. (2004)
pRb$^{+/-}$ and p21$^{Cip1-/-}$		Accelerated development of pituitary tumors	Franklin *et al*. (2000)
pRb$^{+/-}$and p27$^{Kip1-/-}$		Accelerated formation of both pituitary and thyroid tumors	Park *et al*. (1999)
	pRb$^{+/-}$ and E2F1$^{-/-}$	Reduced frequency of pituitary and thyroid tumors	Yamasaki *et al*. (1998)
	pRb$^{+/-}$ and E2F4$^{-/-}$	Reduced frequency of pituitary and thyroid tumors	Lee *et al*. (2002)
E2F1$^{+/-}$ or E2F1$^{-/-}$		Reproductive track sarcomas, lung tumors, lymphomas.	Yamasaki *et al*. (1996); Cloud *et al*. (2002)
p16$^{INK4a-/-}$		Soft tissue sarcomas, splenic lymphomas, melanomas	Sharpless *et al*. (2001, 2004)
p15$^{INK4b-/-}$		Low incidence of hemangiocarcinomas and angiosarcomas	Latres *et al*. (2000)
p18$^{INK4c-/-}$		Pituitary adenomas, adrenal and thyroid tumors	Franklin *et al*. (1998, 2000); Latres *et al*. (2000)
p15$^{INK4b-/-}$and p18$^{INK4c-/-}$		Pituitary adenomas and other tumors characteristic for p15$^{INK4b-/-}$ and p18$^{INK4c-/-}$ mice	Latres *et al*. (2000)
p16$^{INK4a-/-}$and p19$^{ARF-/-}$		Lymphomas, sarcomas, carcinomas	Serrano *et al*. (1996)
p18$^{INK4c-/-}$ and p19$^{INK4d-/-}$		Tumors similar to observed in p18$^{INK4c-/-}$ mice	Zindy *et al*. (2001)
p18$^{INK4c-/-}$ and p21$^{cip1-/-}$		Accelerated development of pituitary tumors	Franklin *et al*. (2000)
p18$^{INK4c-/-}$ and p27$^{Kip1-/-}$		Accelerated development of pituitary tumors	Franklin *et al*. (1998)
p21$^{Cip1-/-}$		Aging animals developed histiocytic sarcomas, hemangiomas, B-cell lymphomas, lung carcinomas	Martin-Caballero *et al*. (2001)
		Accelerated development of mammary and salivary gland tumors induced by Ras	Adnane *et al*. (2000)
p27$^{Kip1-/-}$		Pituitary tumors of the intermediate lobe. Increased sensitivity to tumorigenesis induced by γ-radiation and chemical carcinogenesis in p27$^{Kip1-/-}$ and p27$^{Kip1+/-}$ mice. Increased susceptibility to c-ErbB-2 driven breast cancer in p27$^{Kip1+/-}$ females	Fero *et al*. (1996); Kiyokawa *et al*. (1996); Nakayama *et al*. (1996); Muraoka *et al*. (2002)
p27$^{Kip1-/-}$		Reduced incidence of mammary carcinomas triggered by c-ErbB-2	Muraoka *et al*. (2002)

cells in the auditory epithelium (Chen *et al*., 2003). Unlike mice lacking other INK proteins, p19^{INK4d}-deficient animals did not spontaneously develop tumors (Zindy *et al*., 2000). Ablation of p19^{INK4d} in p18^{INK4c}-null background did not significantly increase the rate of tumor incidence observed in p18^{INK4c} mice (Zindy *et al*., 2001).

As stated above, INK inhibitors are believed to act by disrupting cyclin D–CDK4/6 complexes (reviewed by Ortega *et al*., 2002). Consistent with this model, knockin mice engineered to express CDK4 mutant that is insensitive to INK inhibitors developed a wide spectrum of tumors that cause death between 8 and 16 months of age (Sotillo *et al*., 2001).

The second group of cell cycle inhibitors (Kip/Cip family) consists of p21^{Cip1}, p27^{Kip1} and p57^{Kip2} proteins, which inhibit CDK2-containing complexes (Harper *et al*., 1993; Polyak *et al*., 1994; Lee *et al*., 1995). Moreover, p27^{Kip1} plays a role in inhibiting cyclin–CDK1 complexes (Toyoshima and Hunter, 1994; Nakayama *et al*., 2004; Pagano, 2004).

Although p21^{Cip1} is a p53-inducible gene (el-Deiry *et al*., 1993; Dulic *et al*., 1994; Li *et al*., 1994), during embryonic development its expression is p53-independent. Instead, p21^{Cip1} upregulation correlates with terminal differentiation of a variety of developing tissues such as skeletal and heart muscle, as well as cartilage and skin (Parker *et al*., 1995; Skapek *et al*., 1995; Poolman *et al*., 1998). The fact that p21^{Cip1} is a transcriptional target of p53 suggested a possible role for p21^{Cip1} as a tumor suppressor. This was tested by generation and analyses of mice lacking p21^{Cip1}. p21^{Cip1}-null mice developed normally and were tumor free for the first 7 months of their lives (Deng *et al*., 1995). However, with age they started to develop tumors of hematopoietic, endothelial and epithelial origin, albeit at a rate lower than that seen in p53$^{-/-}$ mice (Martin-Caballero *et al*., 2001). Importantly, absence of p21^{Cip1} accelerated the development of pituitary tumors in pRb$^{+/-}$ mice as well as in animals lacking p18^{INK4c}, suggesting functional collaboration between these proteins in tumor suppression (Brugarolas *et al*., 1998; Franklin *et al*., 2000). Likewise, loss of p21^{Cip1} greatly accelerated tumor incidence in transgenic MMTV-v-Ha-Ras mice expressing oncogenic Ras in their mammary epithelium and in salivary glands; p21$^{Cip1-/-}$/MMTV-v-Ha-Ras animals developed mammary gland and salivary tumors much earlier and with higher incidence than control p21$^{Cip1+/+}$/MMTV-v-Ha-Ras mice (Adnane *et al*., 2000). As mentioned before, mice lacking cyclin D1 were protected from mammary carcinomas driven by MMTV-v-Ha-Ras (Yu *et al*., 2001), underscoring opposite roles of p21^{Cip1} and cyclin D1 in tumorigenesis.

Ablation of p27^{Kip1} did not result in severe developmental defects. However, mutant mice were significantly larger then the control animals (Fero *et al*., 1996; Kiyokawa *et al*., 1996; Nakayama *et al*., 1996). Their internal organs were greatly increased in size, suggesting impaired control of cell proliferation. p27^{Kip1}-deficient females were sterile, most likely due to defects in cell cycle exit during maturation of ovarian corpora lutea (Tong *et al*., 1998). p27$^{Kip1-/-}$ mice also displayed focal retinal abnormalities (Fero *et al*., 1996; Nakayama *et al*., 1996). Moreover, similarly to mice heterozygous for pRb mutation, p27^{Kip1}-deficient animals developed pituitary tumors of the intermediate lobe (Fero *et al*., 1996; Kiyokawa *et al*., 1996; Nakayama *et al*., 1996). Importantly, p27^{Kip1} was shown to be haploinsufficient, as evidenced by the observation that p27$^{Kip1+/-}$ heterozygotes (along with p27$^{Kip1-/-}$ mice) displayed increased sensitivity to tumorigenesis induced by γ-irradiation or by the chemical carcinogens (Fero *et al*., 1998). Likewise, p27$^{Kip1+/-}$ heterozygous females displayed increased susceptibility to mammary carcinomas triggered by the *c-ErbB-2* oncogene. In contrast, p27^{Kip1}-null females had reduced incidence of c-ErbB-2-induced breast tumors (Muraoka *et al*., 2002). This behavior of p27^{Kip1}-deficient animals resembles that of cyclin D1$^{-/-}$ mice (Yu *et al*., 2001), and is consistent with the critical function of p27^{Kip1} in the assembly of cyclin D–CDK complexes (see below).

One of the functions of cyclin D–CDK4/6 complexes in cell cycle progression is to titrate p27^{Kip1} and p21^{Cip1} inhibitors from cyclin E–CDK2 complexes, thereby triggering the kinase activity of cyclin E–CDK2 holoenzyme (Sherr and Roberts, 1999). At the same time p27^{Kip1} and p21^{Kip1} serve as assembly factors for cyclin D-CDK4/6 complexes. A dramatic evidence supporting this model came from the observation that mouse embryonic fibroblasts lacking p21^{Cip1} and p27^{Kip1} proliferated with essentially no cyclin D–CDK4/6 activity (Cheng *et al*., 1999). Also, loss of p27^{Kip1} was shown to correct cell cycle defects seen in CDK4-deficient mouse embryonic fibroblasts (Tsutsui *et al*., 1999). A genetic evidence for cyclin D–p27^{Kip1} interaction was provided by the observation that ablation of p27^{Kip1} in cyclin D1-null background corrected the phenotypes of cyclin D1-deficient mice (Geng *et al*., 2001; Tong and Pollard, 2001).

Mice lacking both p21^{Cip1} and p27^{Kip1} were viable. Gigantism of p27$^{Kip1-/-}$ animals was not increased in double mutant mice, suggesting that these two proteins do not cooperate in the regulation of the body size. However, enlargement of the ovaries, previously observed in p27$^{Kip1-/-}$ females, was even more pronounced in double knockout mice. p21$^{Cip1-/-}$p27$^{Kip1-/-}$females, similarly to p27$^{Kip1-/-}$ females, were sterile due to cell cycle exit defects in cells of the corpora lutea (Jirawatnotai *et al*., 2003).

The amount of cellular p27^{Kip1} is controlled mainly at the level of translation and protein turnover (Hengst and Reed, 1996). Phosphorylation of p27^{Kip1} at threonine 187 by cyclin E–CDK2 complexes targets p27^{Kip1} for ubiquitination and subsequent degradation (Pagano *et al*., 1995; Sheaff *et al*., 1997; Carrano *et al*., 1999; Sutterluty *et al*., 1999; Tsvetkov *et al*., 1999). This pathway has been tested *in vivo* by generation of knockin p27^{T187A} mice, expressing mutated, nondegradable form of p27^{Kip1} (Malek *et al*., 2001). Surprisingly, these mice developed normally and were viable. Analyses of cells derived from the knockin mice led to the discovery of a novel, mitogen-activated proteolytic pathway that governs degradation of $p27^{Kip1}$ in the G1 phase (Malek *et al*., 2001).

Ablation of the least-studied member of the Kip/Cip family, $p57^{Kip2}$, caused very severe developmental abnormalities. Mice lacking $p57^{Kip2}$ died after birth and displayed cleft palate and proliferative disorders in the limbs, lens and intestines that resulted in organomegaly (Yan *et al*., 1997; Zhang *et al*., 1997). These phenotypes resemble the appearance of patients with Beckwith–Wiedemann syndrome. This disease is associated with disruption of genomic imprinting of genes localized on human chromosome 11p15.5, including $p57^{Kip2}$ (Hatada *et al*., 1996; Matsuoka *et al*., 1996). Maternally inherited loss-of-function mutations of $p57^{Kip2}$ have been identified in 10–20% of sporadic Beckwith–Wiedemann syndrome cases (reviewed by Maher and Reik, 2000).

Deletion of both $p57^{Kip2}$ and $p27^{Kip1}$ accelerated the lethality, as double knockout mice died between days 12 and 16.5 of embryonic development. All defects encountered in these double knockout embryos resembled the ones observed in $p57^{Kip2-/-}$ animals. In addition, ablation of both genes caused defective cell cycle exit and defective differentiation of lens fiber cells (Zhang *et al*., 1998). Moreover, loss of $p27^{Kip1}$ significantly increased proliferation rate observed in the labyrinth zone of $p57^{Kip2}$-null placentas. The latter two observations demonstrated that $p27^{Kip1}$ and $p57^{Kip2}$ cooperate in the control of cell cycle exit and differentiation (Zhang *et al*., 1998).

Concluding remarks

Analyses of mice lacking particular cell cycle components revealed that many of the proteins deemed ‘essential’ for cell cycle progression are in fact largely dispensable for mouse development. In certain compartments, however, each of these proteins was shown to play a critical role. Hence, one of the emerging lessons from the mouse knockout experiments is that the cell cycle machinery operates differently in distinct cell types.

In several cases, analyses of the mouse phenotypes provided genetic evidence of interactions between particular cell cycle proteins. For example, the phenotype of mice lacking D-type cyclins closely resembles the appearance of animals lacking cyclin D’s catalytic partners, CDK4 and CDK6 (Kozar *et al*., 2004; Malumbres *et al*., 2004). Combined ablation of p107 and p130 led to more pronounced phenotypes than deletion of p107 or p130 alone, revealing overlapping functions for these proteins in development (Cobrinik *et al*., 1996). Ablation of $p27^{Kip1}$ in cyclin D1 background rescued the phenotypic manifestations of cyclin D1 deficiency (Geng *et al*., 2001; Tong and Pollard, 2001), while ablation of $p27^{Kip1}$ in CDK4-null MEFs corrected the proliferative phenotype of CDK4-deficient cells (Tsutsui *et al*., 1999), thereby providing a genetic evidence for the interaction between cyclin D1–CDK4 and $p27^{Kip1}$. Likewise, ablation of E2Fs or Id2 or N-Ras rescued some of the phenotypes of pRb deficiency (Tsai *et al*., 1998; Lasorella *et al*., 2000; Ziebold *et al*., 2001; Saavedra *et al*., 2002; Takahashi *et al*., 2003). One of the most unexpected outcomes of the knockout analyses is the extremely mild phenotype of mice lacking CDK2 (but not E-type cyclins) (Berthet *et al*., 2003; Geng *et al*., 2003; Ortega *et al*., 2003; Parisi *et al*., 2003), raising a possibility that these cyclins may activate other kinase partners, or perhaps act in a kinase-independent fashion.

In case of several known tumor suppressors (like the retinoblastoma protein), analyses of knockout mice confirmed their tumor-suppressor function, while providing us with very important clues how these proteins operate to constrain cell proliferation. Analyses of mice lacking E2F1 revealed very unexpectedly that this protein–but not other E2Fs, can function as a tumor suppressor (Yamasaki *et al*., 1996). Studies of other E2F knockouts, in turn, provided a wealth of knowledge about the functions of these proteins (Field *et al*., 1996; Yamasaki *et al*., 1996; Humbert *et al*., 2000a, b; Murga *et al*., 2001; Wu *et al*., 2001; Storre *et al*., 2002; Iglesias *et al*., 2004).

While often informative, mouse knockout experiments have several serious limitations. First, analyses of mice lacking a particular protein can only uncover those very selected functions that cannot be performed by other proteins. For example, normal heart development in mice lacking CDK2 does not indicate that CDK2 plays no role in this organ, but only reveals that the functions of CDK2 can also be carried out by other proteins. Along these lines, it is important to note that the ability of two (or more) proteins to substitute for each other does not mean that they normally perform identical functions. For instance, the work of Chen and Pollard (2003) elegantly demonstrated that in uterine epithelial cells, treatment with 17β-estradiol causes translocation of cyclin D1–CDK4 complexes from the cytoplasm to the nucleus. However, in cyclin $D1^{-/-}$ uterine epithelial cells, cyclin D2 acquires cyclin D1’s ability to shuttle between the cytoplasm and the nucleus, thereby activating CDK4 kinase activity, but does so slightly less efficiently than cyclin D1 (Chen and Pollard, 2003).

Another serious limitation of the ‘conventional’ knockout experiments is that the mutant animals develop from the very beginning in the absence of a given protein. Hence, the developing embryo may activate compensatory mechanisms that allow cell proliferation in the absence of a given cell cycle regulator. For instance, Sage *et al*. (2003) demonstrated that cells lacking the retinoblastoma protein upregulate p107, which likely compensates for the loss of pRb. This particular limitation can be overcome by the use of conditional knockout strains that allow a sudden loss of a given protein in mouse tissues or in *in vitro* cultured cells. Indeed, an acute loss of pRb in primary cells was shown to have different phenotypic consequences from germ-line ablation of pRb function (Sage *et al*., 2003).

Despite all these limitations, genetic analyses of mouse development, discussed here, together with genetic studies in other organisms (*Drosophila melanogaster*, *Caenorhabditis elegans*) have played an important role in elucidating the function of the cell cycle machinery. Conditional targeting of the cell cycle components, together with more refined ways of germ-line manipulation that allow introduction of subtle changes into the mouse genome, will undoubtedly continue to contribute to our understanding of the cell cycle machinery for many years to come.

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